

AI PROJECT ESTIMATE & PROPOSAL

Project Scope: uni lms

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Business Domain:	Education
Target Platforms:	Web
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Executive Summary

The proposed project 'uni lms' is designed to serve as a high-quality education platform. Built for web, it incorporates core features including auth, notifications, dashboard, chat. The proposed EdTech platform delivers a virtual learning environment for students and teachers. It emphasizes course management, content consumption, and interactive quizzes. The solution is optimized for varying network conditions, allowing students to access videos and text materials with minimum buffer times.

Cost & Effort Summary

Total Est. Effort	Est. Development Cost	Project Timeline	Team Size
457 hours	\$45,700.00	6-8 weeks	3 members

Recommended Project Roles: Frontend Developer, Backend Developer, QA Engineer / PM

Scope of Work Features

The scope of build covers: Auth, Notifications, Dashboard, Chat, Search, Upload, Ai, Collaboration, Export, Streaming.

Recommended Architecture Stack

Layer	Recommended Technology Stack
Frontend	React + Vite (Tailwind CSS, Framer Motion)
Backend	FastAPI (Python), WebSockets for real-time quiz updates and chat rooms.
Database	MongoDB or PostgreSQL with JSON fields for course curricula.
Hosting & Infrastructure	Cloudflare CDN for video caching, AWS S3 for media storage, AWS Amplify.
External Integrations	Zoom API for live lectures, Vimeo API for video hosting, Zoom SDK for inside-app calls.

Project Implementation Roadmap

Phase / Milestone	Deliverables Scope	Est. Duration
Phase 1: Foundations & User Core	- Database Schema setup - Core API configurations - User Accounts & Authentication module	2 Weeks
Phase 2: Product Core Modules	- Notifications module development - Dashboard module development - Chat module development - Search module development - Upload module development - AI module development - Collaboration module development - Export module development - Streaming module development	3 Weeks
Phase 3: QA, Security & Deployment	- Quality Assurance & Integration tests - Performance optimization runs - Cloud production environment deployment	1 Week

Agile Sprint Planning

Sprint	Sprint Objectives & Effort	Deliverables Checklist
Sprint 1	<i>Objectives:</i> Establish infrastructure, build base APIs, and set up landing dashboard frames. Effort: 228 Hours	• Docker infrastructure and API templates initialized • Database migrations complete and connected locally • User account registration UI & Backend logic verified
Sprint 2	<i>Objectives:</i> Complete end-to-end integration tests, resolve UI issues, and deploy to staging. Effort: 228 Hours	• Full integration unit test suite checks passing • Responsive styling issues resolved across web/mobile viewports • Continuous Integration (CI/CD) pipelines running successfully

Technical Risks & Mitigations

Identified Risk Factor	Proposed Mitigation Strategy
Video streaming buffering on low bandwidth connections	Implement adaptive bitrate streaming (HLS/DASH) and utilize Cloudflare CDN.
Cheating on quizzes and assignments	Randomize question orders, disable copy-paste actions on quiz pages, and add strict submission timeouts.
Media storage costs rising quickly with course uploads	Apply automatic video compression and transcoding tools (AWS Elemental MediaConvert) on upload.